

COURSE TITLE : TEXTILE FIBERS
COURSE CODE : 2061
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	General introduction to Textiles, Fibers & Cotton	15
2	Basic Fibers	15
3	Natural protein Fibers	15
4	Semi – Synthetic Fibers	15
TOTAL		60

COURSE OUTCOME

Module	Go	Student will be able to
1	1	Know about the classification & properties of textile fibres
	2	Understand the cultivation, properties & use of cotton
2	1	Know about the cultivation, properties & use of Linen fibers
	2	Understand the cultivation, properties & use of Jute fibers
3	1	Know the production, processing, properties & uses of Silk
	2	Understand the production, processing, properties & uses of wool
4	1	Know the production, processing, properties & uses of Viscose rayon
	2	Understand the production, processing, properties & uses of Acetate rayon
	3	Know recently developed semi synthetic fibers

SPECIFIC OUTCOME

MODULE I GENERAL INTRODUCTION TO TEXTILES, FIBERS & COTTON

- 1.1.0 **Know about the classification & properties of textile fibres**
 - 1.1.1 State the basic terms related to textiles.
 - 1.1.2 Classify the textile fibers on their origin and chemical nature
 - 1.1.3 Identification of textile fibers.
 - 1.1.4 Explain the essential and desirable properties

1.2.0 Understand the cultivation , properties & use of cotton

- 1.2.1 State the botanical classification of cotton.
- 1.2.2 List the commercial cotton, hybrid Indian cottons and their properties.
- 1.2.3 Define organic cotton & BT cotton.
- 1.2.4 Describe the cultivation of cotton.
- 1.2.5 Illustrate the structure of cotton.
- 1.2.6 State the chemical composition of cotton.
- 1.2.7 List the physical and chemical properties of cotton and its uses

MODULE II BAST FIBERS

2.1.0 Know about the cultivation , properties & use of Linen fibers

- 2.1.1 State the General features of bast fibres
- 2.1.2 Describe the cultivation of linen.
- 2.1.3 Explain the extraction of linen fiber from the plant
- 2.1.4 State the chemical composition of linen
- 2.1.5 Draw the microscopic appearance of linen fiber
- 2.1.6 List the Physical and Chemical properties of linen and its uses

2.2.0 Understand the cultivation , properties & use of Jute fibers

- 2.2.1 Explain the cultivation of jute
- 2.2.2 Describe the extraction of jute fibre from the plant
- 2.2.3 State the chemical composition of jute
- 2.2.4 Draw the microscopic appearance of Jute fibre
- 2.2.5 List the Physical and Chemical properties of Jute and its uses
- 2.2.6 Compare linen with Jute

MODULE III NATURAL PROTEIN FIBERS

3.1.0 Know the production , processing ,properties & uses of Silk

- 3.1.1 List the varieties of Silk and Silk producing countries
- 3.1.2 Explain Sericulture & life cycle of silk worm
- 3.1.3 List the Chemical Composition of raw silk
- 3.1.4 Describe the preparation of Raw silk from cocoons.
- 3.1.5 Explain the method of degumming and weighting of silk
- 3.1.6 List the Physical and Chemical properties & uses of silk
- 3.1.7 Illustrate the structure & microscopic appearance of raw & degummed silk

3.2.0 Understand the production, processing , properties & uses of wool

- 3.2.1 List the varieties of wool and their properties
- 3.2.2 Define the term shearing of wool
- 3.2.3 State the chemical composition of wool
- 3.2.4 Draw the structure & microscopic appearance of wool
- 3.2.5 Explain the scouring, chlorination, and carbonization of wool
- 3.2.6 Define the felting of wool
- 3.2.7 List the physical, chemical properties and uses of wool.

MODULE IV SEMI – SYNTHETIC FIBERS

4.1.0 Know about the production , processing ,properties & uses of Viscose rayon.

- 4.1.1 List the names of semi synthetic fibres
- 4.1.2 Describe the manufacture of Viscose Rayon with a flow chart
- 4.1.3 List the properties and uses of Viscose Rayon
- 4.1.4 Draw the microscopic appearance of the viscose rayon
- 4.1.5 State the salient features of Polynosic Fibres.

4.2.0 Understand the production , processing , properties & uses of Acetate rayon.

- 4.2.1 Describe the method of manufacture of Acetate Rayon
- 4.2.2 List the properties and uses of Acetate Rayon
- 4.2.3 Draw the microscopic appearance of Acetate Rayon

4.3.0 Know recently developed semi synthetic fibers.

- 4.3.1 Define Cuprammonium Rayon, Modal, Lyocell,
- 4.3.2 State FR viscose & Bamboo fibre

CONTENT DETAILS

MODULE I

INTRODUCTION TO TEXTILE FIBRES & COTTON

Definitions of textile fibre, spun yarn, filament yarn, monofilament, texturised yarn, single yarn, ply yarn, etc. - Classification of textile fibres on the basis of origin and chemical nature with examples, General properties of Textile fibre, Essential properties, Desirable properties, Introduction to cotton fibre and cotton producing countries, Botanical classification of cotton, Types of commercial cottons and their properties, Brief study of organic cotton & Bt. cotton, Hybrid Indian Cottons and their properties, Soil conditions, climate, cultivation, Harvesting, picking of cotton, Structure of cotton, longitudinal view and cross sectional view, Chemical compositions, Physical properties & Chemical properties, Uses of Cotton.

MODULE II

BAST FIBRES

Features of Bast fibre like Jute and Linen, Climate, Soil condition and cultivation of linen-Extraction of linen, Rippling, Retting- Chemical, Water and dew retting, Breaking, Scutching, Hackling, Advantages and disadvantages of each retting process, Chemical compositions & microscopic appearance of linen, Physical and chemical properties of linen, cultivation of jute, Extraction of jute fibre from the plant, Chemical and Physical properties of Jute, Importance of Jute in Indian economy, Composition and appearance of jute, Uses of Linen & Jute

MODULE III

NATURAL PROTIEEN FIBRES

Introduction of silk, Verities of silk, silk producing countries, Sericulture, Pre–cocoon operations, Life cycle of silk worm, Various stages involved in production of Raw Silk from cocoons, reeling, Throwing and Doubling of silk, Structure of Silk filaments and appearance, Chemical composition of silk, Degumming and weighting of silk, Physical and chemical properties of silk, Uses of Silk, Introduction to wool and wool producing countries, Varieties of wool, Types& Grades, Shearing of wool, Composition and structure of wool, Chlorination, Carbonisation, Scouring and Felting of wool, Physical and Chemical properties of wool, Uses of wool

MODULE IV

REGENERATED FIBRES

Different types of Rayons, Manufacture of Viscose Rayon with flow chart, Polynosic and properties, Physical and Chemical properties of Viscose Rayon, Uses of Viscose Rayon, Various steps involved in manufacture of acetate Rayon, preparation of dope& spinning with flow chart, Physical and chemical properties of acetate rayon, Uses of acetate Rayon, Microscopical appearance of Viscose & Acetate Rayon, Brief study of cuprammonium and Nitro Rayon, Brief study of Lyocell, Modal, FR viscose & Bamboo fibre.

REFERENCE BOOKS

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|--------------------|--------------------------------------|------------------------------|
| 1. Gohl & Valensky | TEXTILE SCIENCE | Mahajan publishers Ahmadabad |
| 2. J T Marsh | TEXTILE SCIENCE | |
| 3. V.A Shenai | TEXTILE FIBRES | |
| 4. S.P.Mishra | FIBRE SCIENCE AND TECHNOLOGY | |
| 5. Morton & Hearle | PHYSICAL PROPERTIES OF TEXTILE FIBRE | |
| 6. Burno Luniah | IDENTIFICATION OF TEXTILE FIBRES | |